Monte Carlo Option Pricing & Rare-Event Simulation

A Unified Study: Mathematical Foundations, GBM, Heston, CIR, CGMY, Importance Sampling, and Fractional Brownian Motion and Pricing Deep OTM options

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Contents

1 Abstract

This report presents a unified treatment of Monte Carlo simulation and rare-event estimation for stochastic processes in quantitative finance, organised in four parts.

Part I develops the mathematical prerequisites: Monte Carlo integration, the concept of importance sampling, Brownian motion, Itô stochastic calculus, the Euler–Maruyama discretisation scheme, and the Girsanov theorem. These tools form the rigorous foundation on which all subsequent modelling rests, and each is developed from first principles with motivating examples.

Part II introduces the risk-neutral pricing framework and Monte Carlo pricing of European call options under four stochastic models: Geometric Brownian Motion (GBM) / Black-Scholes, the Heston stochastic volatility model, the Cox–Ingersoll–Ross (CIR) interest rate model, and the CGMY Lévy jump process. All models are calibrated to a common slightly OTM contract ($S_0 = 1000$, $K = 1020$, $T = \frac{1}{36}$ yr) and their pricing, distributions, and structure are compared.

Part III presents a deep-learning algorithm for rare-event estimation in diffusions, based on [9]. A neural network approximates the solution to a Hamilton–Jacobi–Bellman (HJB) PDE via a composite loss function and an escalating rarity schedule. Results are validated on GBM and CIR. A diagnostic comparison against simple Monte Carlo on a deep OTM European call quantifies the bias elimination, variance reduction, and wall-clock trade-offs of the NN-IS estimator.

Part IV extends the framework to fractional geometric Brownian motion (fGBM), where Euler–Maruyama breaks down structurally. Four alternative discretisation methods—Milstein, Exact Simulation, Wong–Zakai—are compared under neural-network importance sampling for Hurst indices $H = 0.25$ and $H = 0.75$.

2 Monte Carlo Methods

2.1 What is Monte Carlo Integration?

Many quantities in science and finance can be written as an expectation $\alpha = \mathbb{E}[h(X)]$, where X is a random variable and h is some function of interest. When this expectation cannot be computed analytically, we resort to numerical approximation.

The **Monte Carlo method** approximates α by drawing many independent samples of X and averaging the corresponding values of h.

Definition 2.0.1 (Monte Carlo Estimator)**.** Let $X \sim f$ on $\mathbb{R}^d$ and $h : \mathbb{R}^d \to \mathbb{R}$ be square-integrable, i.e. $\mathbb{E}[h(X)^2] < \infty$. Given N independent samples $X_1, \ldots, X_N \sim f$, the **Monte Carlo estimator** of $\alpha = \mathbb{E}[h(X)]$ is

$$\hat{\alpha}_N \;=\; \frac{1}{N}\sum_{i=1}^{N} h(X_i).$$

The estimator $\hat{\alpha}_N$ is simply the sample average. By the **Law of Large Numbers**, $\hat{\alpha}_N \to \alpha$ almost surely as $N \to \infty$. The key question is: how fast does it converge, and how accurate is it for finite N?

Proposition 2.0.1 (Convergence Rate)**.** *The standard deviation of the Monte Carlo error satisfies*

$$\mathrm{SE}(\hat{\alpha}_N) \;=\; \sqrt{\mathrm{Var}(\hat{\alpha}_N)} \;=\; \frac{\sqrt{\mathrm{Var}(h(X))}}{\sqrt{N}} \;=\; \mathcal{O}(N^{-1/2}).$$

Proof. Since $X_1, \ldots, X_N$ are independent and identically distributed,

$$\mathrm{Var}(\hat{\alpha}_N) = \mathrm{Var}\Big(\frac{1}{N}\sum_{i=1}^{N} h(X_i)\Big) = \frac{1}{N^2}\sum_{i=1}^{N} \mathrm{Var}(h(X_i)) = \frac{\mathrm{Var}(h(X))}{N}.$$

Taking square roots gives $\mathrm{SE}(\hat{\alpha}_N) = \sigma_h/\sqrt{N}$, where $\sigma_h^2 = \mathrm{Var}(h(X))$. □

The $\mathcal{O}(N^{-1/2})$ rate is *dimension-free*: it does not depend on the dimension d of the problem. This makes Monte Carlo uniquely powerful in high dimensions, where deterministic methods (finite differences, quadrature) require a number of grid points that grows exponentially in d.

2.2 The Problem of Rare Events

Suppose we want to estimate a small probability $p = \mathbb{P}(X \geq B)$ for a large threshold B. With $h(x) = \mathbf{1}_{x \geq B}$, the MC estimator is just the fraction of samples that exceed B. The relative error of the estimator is

$$\frac{\mathrm{SE}(\hat{p})}{p} = \frac{\sqrt{p(1-p)/N}}{p} \approx \frac{1}{\sqrt{Np}}.$$

The fundamental problem is that rare events are almost never seen in simulation, so you're estimating a probability from almost no actual occurrences. More formally, the number of "hits" you expect is just Np, and when $Np \ll 1$, you might see zero hits in your entire simulation. Importance sampling (Section 3) is the standard remedy.

3 Importance Sampling

3.1 The Core Idea

The problem with plain Monte Carlo for rare events is that most samples contribute nothing—they fall in the common region where $h(x) \approx 0$, while the rare region where $h(x)$ is large is almost never visited. **Importance sampling** fixes this by deliberately biasing the simulation toward the region that matters, and then correcting for the bias using a likelihood ratio weight.

Definition 3.0.1 (Importance Sampling Estimator)**.** Let g be any density satisfying $f(x) > 0 \Rightarrow g(x) > 0$ (so g covers the support of f). Rewrite the expectation as

$$\alpha = \mathbb{E}_f[h(X)] = \int h(x) f(x)\, dx = \int h(x) \frac{f(x)}{g(x)} g(x)\, dx = \mathbb{E}_g\Big[h(\tilde{X})\, \frac{f(\tilde{X})}{g(\tilde{X})}\Big].$$

For $\tilde{X}_1, \ldots, \tilde{X}_N \sim g$, the **IS estimator** is

$$\hat{\alpha}_g = \frac{1}{N} \sum_{i=1}^{N} h(\tilde{X}_i)\, w(\tilde{X}_i), \qquad w(x) = \frac{f(x)}{g(x)}.$$

The factor $w(x)$ is called the **likelihood ratio** or **Radon-Nikodym derivative** of f with respect to g.

The estimator $\hat{\alpha}_g$ is *unbiased*: $\mathbb{E}_g[\hat{\alpha}_g] = \alpha$ for any valid choice of g. The variance under g is

$$\mathrm{Var}_g(\hat{\alpha}_g) = \frac{1}{N} \Big(\mathbb{E}_g[h(\tilde{X})^2 w(\tilde{X})^2] - \alpha^2 \Big).$$

IS reduces variance when g concentrates mass where $|h(x)|w(x)$ is small—intuitively, when g is "shaped like" $|h(x)|f(x)$.

3.2 The Zero-Variance Estimator

Proposition 3.0.1. *The choice $g^*(x) = |h(x)|f(x)/\alpha$ achieves zero variance:* $\mathrm{Var}_{g^*}(\hat{\alpha}_{g^*}) = 0$.

Proof. Under g^*, every sample gives $h(\tilde{X}_i)w(\tilde{X}_i) = h(\tilde{X}_i) \cdot f(\tilde{X}_i)/g^*(\tilde{X}_i) = \alpha$ deterministically. Hence all terms in the sum equal α and the variance is zero. $\square$

Of course, g^* requires knowledge of α—the very quantity we seek—so it cannot be used directly. However, it gives a concrete target: the closer g is to g^*, the lower the variance. For diffusion processes, finding the best g reduces to solving an optimal control problem (see Section 9).

Example 3.0.1 (Tail probability via IS)**.** Let $X \sim \mathcal{N}(0, 1)$ under $\mathbb{P}$ and consider $\alpha = \mathbb{P}(X \geq A)$ for large A. Plain MC gives relative error $\approx 1/\sqrt{N\Phi(-A)}$, which is enormous for large A. Under the shifted measure $\tilde{X} \sim \mathcal{N}(A, 1)$ (i.e. $g(x) = \phi(x - A)$), the probability that $\tilde{X} \geq A$ is $\frac{1}{2}$, so roughly half the samples are useful. The likelihood ratio is $w(x) = \phi(x)/\phi(x - A) = e^{-Ax + A^2/2}$, which corrects for the shift. This gives a dramatically more efficient estimator, especially for large A.

4 Stochastic Differential Equations

4.1 Brownian Motion

Classical calculus deals with smooth functions. Asset prices, however, fluctuate erratically. The mathematical model for pure randomness in continuous time is Brownian motion (also called the Wiener process).

Definition 4.0.1 (Standard Brownian Motion)**.** A stochastic process $\{W_t\}_{t\geq 0}$ is a **standard Brownian motion** if:

1. $W_0 = 0$ almost surely.
2. **Independent increments**: for $0 \leq s < t \leq u < v$, the increments $W_t - W_s$ and $W_v - W_u$ are independent.
3. **Gaussian increments**: $W_t - W_s \sim \mathcal{N}(0, t-s)$ for all $0 \leq s < t$.
4. **Continuous paths**: $t \mapsto W_t$ is continuous almost surely.

Several consequences are immediate. The variance of an increment over time interval $[s,t]$ is $t-s$, so longer intervals give more uncertainty. Brownian motion is nowhere differentiable—its paths are too irregular for classical derivatives. This forces us to develop a new kind of calculus.

4.2 Itô Stochastic Integrals and SDEs

An SDE models a system whose rate of change depends both on time and on random fluctuations driven by Brownian motion.

Definition 4.0.2 (Itô SDE)**.** A **stochastic differential equation** in Itô form is

$$dX_t = \mu(t, X_t)\,dt + \sigma(t, X_t)\,dW_t, \quad X_0 = x_0.$$

Here $\mu(t,x)$ is the **drift** (the average rate of change) and $\sigma(t,x)$ is the **diffusion coefficient** (scaling the random noise). In integral form:

$$X_t = x_0 + \int_0^t \mu(s, X_s)\,ds + \int_0^t \sigma(s, X_s)\,dW_s.$$

The first integral is an ordinary Lebesgue integral; the second is an Itô stochastic integral.

Under Lipschitz and linear growth conditions on μ and σ, the SDE has a unique strong solution (Existence and Uniqueness Theorem).

4.3 Itô's Lemma: The Chain Rule for SDEs

In ordinary calculus, if $Y_t = f(X_t)$ and X_t is differentiable, then $dY_t = f'(X_t)\,dX_t$. For stochastic processes, Brownian motion has non-zero quadratic variation ($dW_t \cdot dW_t = dt$), which adds an extra correction term.

Theorem 4.1 (Itô's Lemma)**.** *Let X_t satisfy $dX_t = \mu_t\,dt + \sigma_t\,dW_t$ and let $f(t,x)$ be twice continuously differentiable. Then $Y_t = f(t, X_t)$ satisfies*

$$dY_t = \underbrace{\partial_t f\,dt}_{\textit{time change}} + \underbrace{\partial_x f\,dX_t}_{\textit{chain rule}} + \underbrace{\frac{1}{2}\sigma_t^2\,\partial_{xx} f\,dt}_{\textit{Itô correction}}.$$

Written out in full:

$$dY_t = \left(\partial_t f + \mu_t\,\partial_x f + \tfrac{1}{2}\sigma_t^2\,\partial_{xx} f\right) dt + \sigma_t\,\partial_x f\,dW_t.$$

The extra term $\frac{1}{2}\sigma_t^2\,\partial_{xx}f\,dt$ is the **Itô correction**. It arises because $(dW_t)^2 = dt \neq 0$, unlike in ordinary calculus where $(dx)^2 = 0$.

Example 4.1.1 (Solving GBM via Itô's Lemma)**.** Consider $dS_t = rS_t\,dt + \sigma S_t\,dW_t$ (Geometric Brownian Motion). We want to find S_T explicitly. Let $f(S) = \ln S$, so $\partial_x f = 1/S$ and $\partial_{xx} f = -1/S^2$. By Itô's lemma:

$$\begin{aligned} d(\ln S_t) &= \frac{1}{S_t}\,dS_t - \frac{1}{2}\cdot\frac{1}{S_t^2}\cdot\sigma^2 S_t^2\,dt \\ &= \frac{1}{S_t}(rS_t\,dt + \sigma S_t\,dW_t) - \frac{\sigma^2}{2}\,dt \\ &= \Big(r - \frac{\sigma^2}{2}\Big)dt + \sigma\,dW_t. \end{aligned}$$

This is now an SDE with *constant* coefficients, which integrates directly:

$$\ln S_T - \ln S_0 = \Big(r - \frac{\sigma^2}{2}\Big)T + \sigma W_T.$$

Exponentiating:

$$\boxed{S_T = S_0\,\exp\Big[\Big(r - \frac{\sigma^2}{2}\Big)T + \sigma\sqrt{T}\,Z\Big], \quad Z \sim \mathcal{N}(0,1).}$$

The term $-\sigma^2/2$ (the Itô correction) ensures that $\mathbb{E}[S_T] = S_0 e^{rT}$. Without it, the expected stock price would grow at the wrong rate.

4.4 Euler–Maruyama Discretisation

When an SDE has no known closed-form solution (as with Heston or CIR), we simulate it numerically by discretising time into small steps.

Definition 4.1.1 (Euler–Maruyama Scheme)**.** Partition $[0,T]$ into m steps of size $\Delta t = T/m$. Starting from $Y_0 = X_0$, update iteratively:

$$Y_{j+1} = Y_j + \mu(t_j, Y_j)\,\Delta t + \sigma(t_j, Y_j)\,\sqrt{\Delta t}\,Z_j, \quad Z_j \overset{\text{iid}}{\sim} \mathcal{N}(0,1).$$

At each step, the drift and diffusion coefficients are *frozen* at their current values, and the Brownian increment $\Delta W_j = \sqrt{\Delta t}\,Z_j$ is simulated exactly.

This is the stochastic analogue of Euler's method for ODEs. Its accuracy improves as $\Delta t \to 0$:

4.5 Girsanov's Theorem: Changing the Drift

Girsanov's theorem provides the conditions under which a change of probability measure transforms a Brownian motion with drift into a standard Brownian motion, thereby enabling a controlled change of drift in stochastic differential equations.

Intuition. Suppose under measure $\mathbb{P}$ a Brownian motion W_t has no drift, and we want to simulate under a new measure $\mathbb{Q}$ where W_t behaves like a Brownian motion with drift b. Girsanov's theorem says this is possible: define a new process $W_t^{\mathbb{Q}} = W_t - \int_0^t b(s)\,ds$, and there exists a measure $\mathbb{Q}$ equivalent to $\mathbb{P}$ under which $W_t^{\mathbb{Q}}$ is a standard Brownian motion.

Theorem 4.2 (Girsanov's Theorem)**.** *Let W_t be a standard Brownian motion under $\mathbb{P}$ and let $b = \{b(t)\}$ be an adapted process satisfying* ***Novikov's condition****:*

$$\mathbb{E}^{\mathbb{P}}\Big[\exp\Big(\frac{1}{2}\int_0^T b(s)^2\,ds\Big)\Big] < \infty.$$

*Define the **Girsanov exponential**:*

$$\mathcal{E}_T(b) = \exp\Bigl(-\frac{1}{2}\int_0^T b(s)^2\,ds + \int_0^T b(s)\,dW_s\Bigr).$$

Then the measure $\mathbb{Q}$ defined by $d\mathbb{Q}/d\mathbb{P} = \mathcal{E}_T(b)$ is equivalent to $\mathbb{P}$, and under $\mathbb{Q}$ the process

$$W_t^{\mathbb{Q}} = W_t - \int_0^t b(s)\,ds$$

is a standard Brownian motion.

Corollary 4.2.1 (Effect on the SDE)**.** *Suppose $dX_t = \mu(t, X_t)\,dt + \sigma(t, X_t)\,dW_t$ under $\mathbb{P}$. Since $dW_t = dW_t^{\mathbb{Q}} + b(t)\,dt$, the same process X_t satisfies*

$$dX_t = \bigl[\mu(t, X_t) + \sigma(t, X_t)\,b(t)\bigr]\,dt + \sigma(t, X_t)\,dW_t^{\mathbb{Q}}$$

under $\mathbb{Q}$. The drift shifts from μ to $\mu + \sigma b$, while σ is unchanged.

This is exactly what importance sampling does in continuous time: by choosing b carefully, we can bias the SDE toward the rare region and then reweight by $d\mathbb{P}/d\mathbb{Q} = 1/\mathcal{E}_T(b)$ to obtain an unbiased estimator.

Example 4.2.1 (IS for a diffusion)**.** To estimate $p = \mathbb{P}(X_T > a)$ where a is a large threshold, we add drift $b > 0$ to push X_t upward. Under the new measure $\mathbb{Q}^b$, paths reach a far more often. The IS estimator is

$$\hat{p} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\bigl\{X_T^{(i)} > a\bigr\} \cdot \frac{d\mathbb{P}}{d\mathbb{Q}^b}\bigg|_{\text{path } i},$$

where the likelihood ratio corrects for the artificial bias. Choosing b optimally is the subject of Section 9.

5 Financial Setup and Risk-Neutral Pricing

5.1 European Call Options

A **European call option** gives its holder the right—but not the obligation—to purchase an underlying asset (e.g. a stock) at a fixed price K (the **strike**) exactly at a future time T (the **maturity**).

Definition 5.0.1 (European Call Option)**.** A European call option with strike K and maturity T has terminal payoff

$$h(S_T) = \max(S_T - K,\, 0) = (S_T - K)^+.$$

The intuition is simple: if at maturity the stock price $S_T > K$, the holder buys at K and immediately sells at S_T, earning a profit of $S_T - K$. If $S_T \leq K$, the holder simply does not exercise, and the payoff is zero.

The option is called **out-of-the-money** (OTM) if $S_0 < K$, meaning the stock would need to rise for the option to be profitable. Its value is therefore a bet on the stock rising sufficiently over T.

5.2 Risk-Neutral Pricing and Monte Carlo

Under the risk-neutral measure $\mathbb{Q}$ (a probability measure under which all assets grow at the risk-free rate r), the fair price of any derivative is its discounted expected payoff:

$$C = e^{-rT}\, \mathbb{E}^{\mathbb{Q}}\big[\max(S_T - K,\, 0)\big].$$

This is the fundamental pricing formula. For most models, this expectation cannot be computed in closed form, and we approximate it by Monte Carlo:

$$\hat{C} = e^{-rT} \cdot \frac{1}{N} \sum_{i=1}^{N} \max(S_T^{(i)} - K,\, 0),$$

where $S_T^{(1)}, \ldots, S_T^{(N)}$ are independent simulated terminal stock prices. By Proposition 2.0.1, $\mathrm{SE}(\hat{C}) = \sigma_P/\sqrt{N}$ where $\sigma_P^2 = \mathrm{Var}(e^{-rT}(S_T - K)^+)$.

5.3 Common Parameters

All models in this report are calibrated to the same contract:

Symbol	Value	Description
S_0	1,000	Initial stock price
K	1,020	Strike price (slightly OTM)
T	$1/36 \approx 0.028\,\mathrm{yr}$	Maturity ($\approx$ 10 calendar days)
r	0.05	Continuously compounded risk-free rate
N	1,000 / 10,000	Paths per run (GBM / advanced models)
M	1,000	Repeated simulation runs (GBM study)

Table 1: Shared financial and simulation parameters across all models.

6 Geometric Brownian Motion and the Black-Scholes Framework

6.1 The GBM Model

The simplest and most widely used model for stock prices is Geometric Brownian Motion. Under the risk-neutral measure:

Definition 6.0.1 (Geometric Brownian Motion)**.**

$$dS_t = r\,S_t\,dt + \sigma\,S_t\,dW_t^{\mathbb{Q}}, \quad S_0 > 0.$$

Here r is the risk-free rate (the drift under $\mathbb{Q}$) and $\sigma > 0$ is the **volatility**—a constant that measures how much the stock fluctuates. The term $\sigma S_t\,dW_t$ says that the random shock to S_t is proportional to S_t itself (percentage shocks are constant), which leads to a log-normal distribution for S_T.

6.2 Exact Solution via Itô's Lemma

We already derived the solution in Example 4.1.1. Applying Itô's lemma to $f(S) = \ln S$:

$$\ln S_T - \ln S_0 = \Big(r - \frac{\sigma^2}{2}\Big)T + \sigma W_T,$$

so

$$\boxed{S_T = S_0\,\exp\Big[\Big(r - \frac{\sigma^2}{2}\Big)T + \sigma\sqrt{T}\,Z\Big], \quad Z \sim \mathcal{N}(0,1).}$$

This means $\ln(S_T/S_0)$ is normally distributed, so S_T is **log-normally distributed**. Key moments:

$$\mathbb{E}^{\mathbb{Q}}[S_T] = S_0 e^{rT}, \qquad \mathrm{Var}^{\mathbb{Q}}(S_T) = S_0^2 e^{2rT}(e^{\sigma^2 T} - 1).$$

The correction $-\sigma^2/2$ in the exponent is the Itô correction: without it, the expected value would be wrong.

6.3 Black-Scholes Closed Form

Since S_T is log-normal, the expectation $\mathbb{E}^{\mathbb{Q}}[(S_T - K)^+]$ can be evaluated analytically.

Proposition 6.0.1 (Black-Scholes Formula, 1973)**.** *Under GBM with constant r and σ:*

$$C_{\mathrm{BS}} = S_0\,\Phi(d_1) - Ke^{-rT}\Phi(d_2),$$

where

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T},$$

and $\Phi(\cdot)$ is the standard normal CDF.

Derivation sketch. Write $C = e^{-rT}\mathbb{E}[(S_T - K)^+]$. Let $\mu_{\ln} = \ln S_0 + (r - \sigma^2/2)T$ and $\sigma_{\ln} = \sigma\sqrt{T}$, so $\ln S_T \sim \mathcal{N}(\mu_{\ln}, \sigma_{\ln}^2)$. Then

$$\begin{aligned}
\mathbb{E}[(S_T - K)^+] &= \int_{-\infty}^{\infty} (e^u - K)^+ \frac{1}{\sigma_{\ln}\sqrt{2\pi}} e^{-(u-\mu_{\ln})^2/(2\sigma_{\ln}^2)}\,du \\
&= \int_{\ln K}^{\infty} (e^u - K) \frac{1}{\sigma_{\ln}\sqrt{2\pi}} e^{-(u-\mu_{\ln})^2/(2\sigma_{\ln}^2)}\,du.
\end{aligned}$$

Substituting $z = (u - \mu_{\ln})/\sigma_{\ln}$ and splitting into two integrals, the first gives $S_0 e^{rT}\Phi(d_1)$ (using the moment generating function of the normal) and the second gives $K\Phi(d_2)$. Discounting by e^{-rT} gives the result. □

6.4 GBM Numerical Results

Metric	Value
Black-Scholes Price C_{BS}	*9.095420*
MC Mean Price $\bar{C}$	*9.047123*
MC Standard Deviation s_M	*18.615668*
95% Confidence Interval	*7.893333 - 10.200913*
Absolute Error $\lvert\bar{C} - C_{\mathrm{BS}}\rvert$	*0.048297*

Table 2: GBM Monte Carlo results vs. Black-Scholes benchmark ($\sigma = 0.25$, $N = 1000$, $M = 1000$).

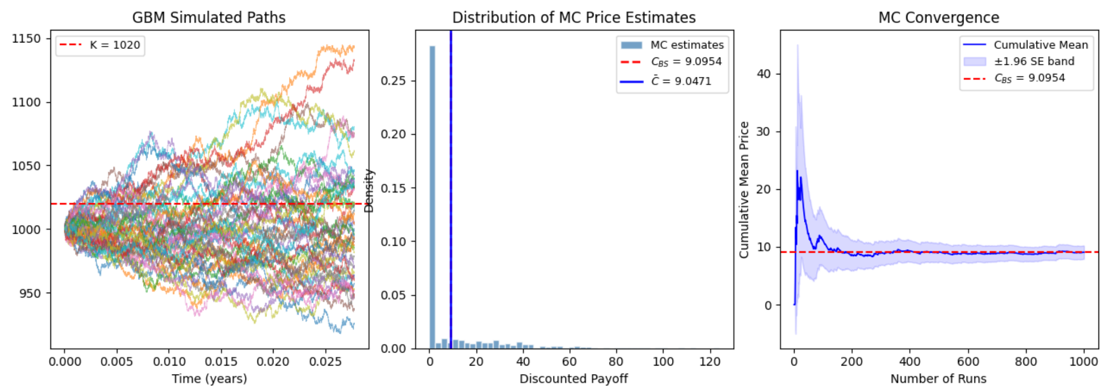

Figure 1: GBM simulation diagnostics ($\sigma = 0.25$, $N = 1000$).

7 Advanced Stochastic Models

7.1 Why Move Beyond GBM?

GBM assumes (i) *constant* volatility, (ii) *continuous* price paths, and (iii) *normally* distributed log-returns. All three are empirically violated:

- **Volatility clustering**: markets go through calm and turbulent periods; volatility is not constant.
- **Implied volatility smile**: if GBM were correct, options at different strikes would imply the same σ. In practice, the implied vol varies with strike—a "smile" or "skew".
- **Heavy tails**: large market crashes happen far more often than the normal distribution predicts.
- **Jumps**: prices can move discontinuously in a crisis (e.g. a company announcement or market crash).

Feature	GBM	Heston	CIR	CGMY
Volatility	Constant	Stochastic	Fixed equity vol	Jump-implicit
Jumps	No	No	No	Yes (infinite)
Path type	Continuous	Continuous	Continuous	Discontinuous
Correlation	—	$\rho(S, v)$	Independent	—
Closed form	BS formula	FFT semi-analytic	Bond pricing	Char. function

Table 3: Key structural characteristics of all four models.

7.2 Heston Stochastic Volatility Model

The key insight of the Heston model (1993) is that volatility itself is random and evolves over time, driven by its own stochastic process.

Definition 7.0.1 (Heston Model)**.** Under $\mathbb{Q}$, the joint dynamics of stock price S_t and its instantaneous variance v_t are:

$$\begin{aligned} dS_t &= r\,S_t\,dt + \sqrt{v_t}\,S_t\,dW_1, \\ dv_t &= \kappa(\theta - v_t)\,dt + \xi\sqrt{v_t}\,dW_2, \end{aligned}$$

with $\mathrm{Corr}(dW_1, dW_2) = \rho\,dt$.

The second equation is itself an SDE: v_t is attracted toward a long-run mean θ at speed κ (mean-reverting), but is also pushed around by noise scaled by $\xi\sqrt{v_t}$.

Parameter interpretation:

- v_0: initial variance (spot volatility $= \sqrt{v_0}$).
- κ: how quickly v_t reverts to θ after a shock.
- θ: the long-run average level of variance.
- ξ: "vol of vol"; how volatile the volatility itself is.
- $\rho < 0$: the **leverage effect**—when the stock falls, volatility tends to rise (negative correlation). This creates the observed left skew in equity returns.

The **Feller condition** $2\kappa\theta > \xi^2$ ensures that v_t remains strictly positive. With $\kappa = 2.0$, $\theta = 0.09$, $\xi = 0.05$: $2 \times 2 \times 0.09 = 0.36 \gg 0.0025 = \xi^2$.

To simulate correlated Brownian increments, we use the Cholesky decomposition: $dW_1 = Z_1\sqrt{\Delta t}$ and $dW_2 = (\rho Z_1 + \sqrt{1-\rho^2}\, Z_2)\sqrt{\Delta t}$, where $Z_1, Z_2 \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$. One can verify $\text{Corr}(dW_1, dW_2) = \rho\, \Delta t$.

Parameters used: $v_0 = 0.09$, $\kappa = 2.0$, $\theta = 0.09$, $\xi = 0.05$, $\rho = -0.5$.

7.3 Cox–Ingersoll–Ross Model

The CIR model (1985) was originally developed to model interest rates that are always positive and revert to a long-run level. Here we use it to model stochastic short rates that drive the stock price.

Definition 7.0.2 (CIR Model)**.** The short rate r_t follows:

$$dr_t = \kappa(\theta - r_t)\, dt + \sigma\sqrt{r_t}\, dW_t^{\mathbb{Q}}.$$

The structure is similar to Heston's variance process. The key properties are:

- **Mean reversion**: r_t is pulled back toward θ at rate κ.
- **Non-negativity**: the $\sqrt{r_t}$ diffusion shrinks as $r_t \to 0$, preventing it from going negative. The Feller condition $2\kappa\theta > \sigma^2$ guarantees strict positivity.
- **Exact transition density**: r_t given r_s follows a non-central chi-squared distribution, enabling exact (not just approximate) simulation.

The asset price under stochastic rate r_t evolves as

$$S_{t+\Delta t} = S_t \exp\bigl[(r_t - \tfrac{1}{2}\sigma_{\text{eq}}^2)\, \Delta t + \sigma_{\text{eq}}\sqrt{\Delta t}\, Z\bigr],$$

where $\sigma_{\text{eq}} = 0.25$ is a fixed equity volatility. Parameters: $r_0 = 0.05$, $\kappa = 2.0$, $\theta = 0.05$, $\sigma = 0.10$. Feller: $0.20 > 0.01$.

7.4 CGMY Lévy Model

The CGMY model (Carr, Geman, Madan, Yor, 2002) allows for *jumps*: sudden, discontinuous movements in the stock price. Rather than continuous Brownian diffusion, the randomness comes from a Lévy process—a process with independent, stationary increments that can have jumps.

Definition 7.0.3 (CGMY Lévy Measure)**.** The CGMY process X_t is characterised by its Lévy measure (which describes the distribution of jump sizes):

$$\nu(dx) = C\frac{e^{-Mx}}{x^{1+Y}}\mathbf{1}_{x>0}\, dx + C\frac{e^{-G|x|}}{|x|^{1+Y}}\mathbf{1}_{x<0}\, dx.$$

Parameter interpretation:

- $C > 0$: overall jump intensity (more jumps per unit time).
- $G > 0$: controls the decay of large *negative* jumps (left tail).
- $M > 0$: controls the decay of large *positive* jumps (right tail).
- $Y \in (0, 2)$: tail heaviness and fine structure; smaller Y means more small jumps and heavier power-law tails.

With $G = M = 5.0$, the left and right tails are symmetric (no skewness from jump sizes). With $Y = 0.5$, the process has infinite activity (infinitely many small jumps) but finite variation.

The risk-neutral log-price is $S_T = S_0 \exp[(r-\omega)T + X_T]$, where $\omega = C\Gamma(-Y)[(M-1)^Y - M^Y + (G+1)^Y - G^Y]$ is the martingale correction that ensures $\mathbb{E}^{\mathbb{Q}}[S_T] = S_0 e^{rT}$.

Simulation: truncate the Lévy measure at small jump size ε, draw the number of large jumps from a Poisson distribution, and sample their sizes from the normalised Lévy measure.

8 Simulation Results and Comparative Analysis

8.1 Cross-Model Pricing Results

Metric	GBM	Heston	CIR	CGMY
Option Price $\hat{C}$	8.9556	11.3075	8.0599	10.0753
Standard Error	0.5691	0.6893	0.5420	1.3373
95% CI lower	7.8402	9.9565	6.9976	7.4542
95% CI upper	10.0710	12.6586	9.1222	12.6963
Mean S_T	1000.91	998.75	998.78	1000.24
Std Dev S_T	41.83	49.79	40.50	62.73
% Paths in-the-money	32.40	34.50	30.10	17.50

Table 4: Monte Carlo pricing results across all four models ($N = 10{,}000$).

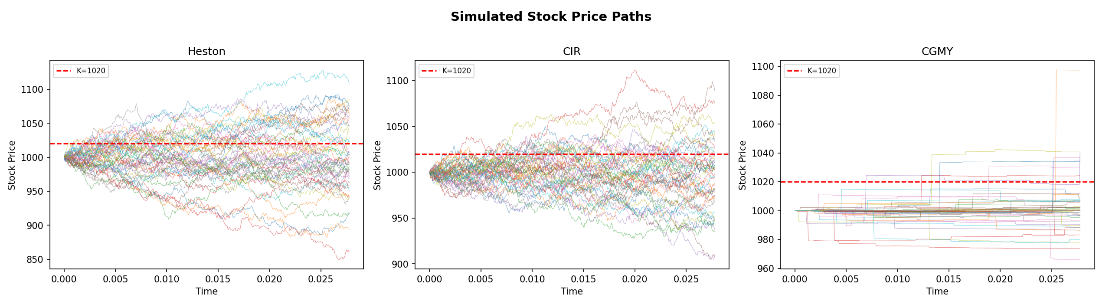

Figure 2: Simulation for Heston, CIR, and CGMY models ($N = 10{,}000$).

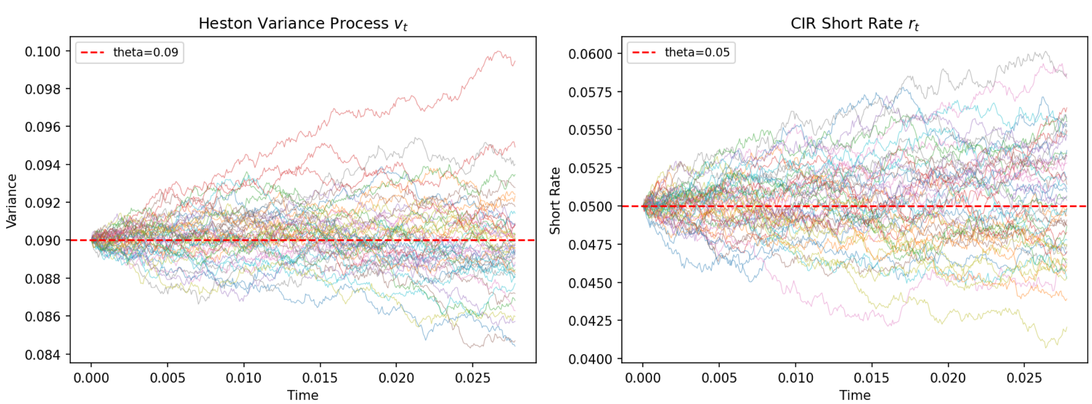

Figure 3: Heston variance v_t and CIR rate r_t auxiliary processes.

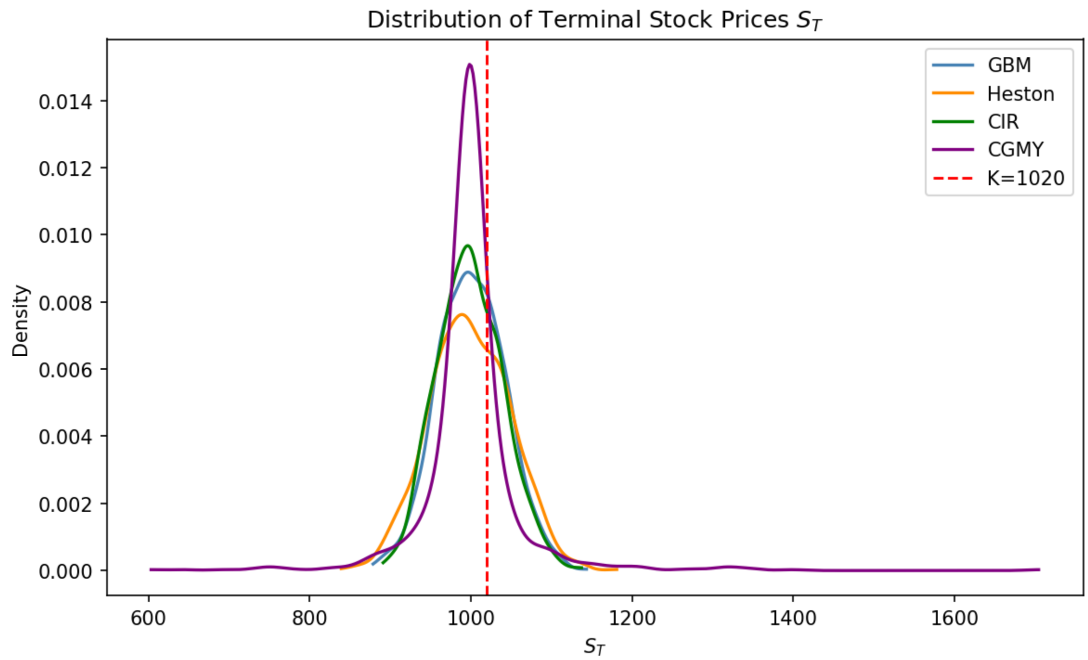

Figure 4: terminal price KDE (all four models) and payoff histograms.

8.2 Pricing Intuition for the OTM Contract

For the 10-day, 2%-OTM contract ($S_0 = 1000$, $K = 1020$):

- **GBM**: The reference price. Over 10 days, the stock must rise by 2% for the option to pay off—a Brownian diffusion problem.
- **Heston**: Stochastic volatility can spike and push paths further out, but the leverage effect ($\rho = -0.5$) slightly reduces the upward tail probability (stock rises tend to coincide with vol decreases). Net effect on the price is modest.
- **CIR**: Stochastic rates modestly affect drift and discounting over 10 days; price is close to GBM at the same equity volatility σ_{eq}.
- **CGMY**: Largest price. A single large upward jump can instantly push S_T well past K, providing a "jump premium" absent from diffusion models. This effect is especially pronounced at short maturities where diffusion contributes little.

Property	Heston / CIR (diffusion)	CGMY (jumps)
Path continuity	Continuous a.s.	Discontinuous (*càdlàg*)
Small-time behaviour	$\sim \sigma_0\sqrt{T}$	Dominated by jump activity
Tail behaviour	Semi-heavy (log-normal)	Heavy (power-law, Y-controlled)
Short-dated OTM	Underpriced vs. market	Better fit
Hedging	Delta-gamma sufficient	Requires jump-risk exposure

Table 5: Diffusion vs. jump process properties.

9 Deep-Learning Importance Sampling for Rare Events

This section presents the algorithm of [9] for estimating rare-event probabilities in diffusion processes using neural networks.

9.1 The Small-Noise SDE and the Rare-Event Problem

Consider the diffusion process

$$dX_t = b(X_t, t)\,dt + \sigma(X_t, t)\sqrt{\varepsilon}\,dB_t, \quad X_0 = x_0,$$

where $\varepsilon > 0$ controls the noise level. We want to estimate the probability $p_\varepsilon = \mathbb{P}(X_T > a)$ for a threshold $a \gg x_0$—a rare event when ε is small.

As $\varepsilon \to 0$, the process becomes deterministic and $p_\varepsilon \to 0$ exponentially fast. Standard MC requires $\mathcal{O}(1/p_\varepsilon)$ samples to get even a single success, making it computationally hopeless for small p_ε.

By the IS argument from Section 3 and Girsanov's theorem (Theorem 4.2), we can add a control drift $u(t, X_t)$ to the SDE:

$$dX_t^u = \left[b + \sigma\sqrt{\varepsilon}\,u\right] dt + \sigma\sqrt{\varepsilon}\,dB_t^u,$$

and estimate p_ε by simulating under this biased dynamics, then reweighting by the Girsanov likelihood ratio. The IS estimator is unbiased for any u, and achieves zero variance at the optimal control u^*.

9.2 HJB PDE for the Optimal Control

To find u^*, we minimise the second moment of the IS estimator. This leads to the value function $V^\varepsilon(t,x) = \inf_u \mathbb{E}[\cdot]$. The log-transform $W^\varepsilon = -\varepsilon \log V^\varepsilon$ (which removes the exponential dependence on ε) satisfies the HJB PDE:

$$-\partial_t W^\varepsilon - b\,\partial_x W^\varepsilon + \frac{\sigma^2(\partial_x W^\varepsilon)^2}{2} - \frac{\varepsilon\sigma^2}{2}\,\partial_{xx} W^\varepsilon = 0, \qquad W^\varepsilon(T,x) = 2f(x), \tag{1}$$

where $f(x) = 0$ for $x \geq a$ and $+\infty$ otherwise. The optimal control is then recovered as $u^* = \frac{\sigma}{2\sqrt{\varepsilon}}\partial_x W^\varepsilon$.

Intuitively, W^ε encodes how "hard" it is to reach the rare region from each point (t,x). Solving (1) gives us the optimal drift to apply at each time-space point.

9.3 Neural Network Approximation

The HJB PDE (1) is generally intractable analytically. We approximate W^ε by a feedforward neural network $W_\theta^\varepsilon(t,x)$ with: input $(t,x) \in \mathbb{R}^2$; two hidden layers of width 5; tanh activations; scalar output. The optimal control is computed by automatic differentiation: $u_\theta = \frac{\sigma\sqrt{\varepsilon}}{2}\partial_x W_\theta^\varepsilon$.

Training minimises the composite loss

$$\mathcal{L}(\theta) = \underbrace{-k_1\, W_\theta(0, x_0)}_{\text{(i) variance min}} + \underbrace{k_2 \int_0^T \tfrac{1}{2}\,\mathbb{E}[\mathcal{L}^\varepsilon W_\theta]^2\,dt}_{\text{(ii) HJB residual}} + \underbrace{k_3\,\mathbb{E}[(W_\theta(T, X_T) - 2f(X_T))^2]}_{\text{(iii) terminal BC}}, \tag{2}$$

with weights $(k_1, k_2, k_3) = (0.35, 0.45, 0.20)$.

Term (i) directly maximises $W(0,x_0) = -\varepsilon \log V^\varepsilon(0, x_0)$, which is equivalent to minimising the second moment of the IS estimator (the primary goal). Term (ii) enforces that W_θ satisfies the HJB PDE along simulated trajectories; it is the dominant term and setting $k_2 = 0$ causes the training to diverge. Term (iii) anchors the terminal condition and stabilises training.

9.4 Escalating Rarity Schedule

Training directly at $\varepsilon = 1$ (the true rare-event problem) is ill-conditioned: virtually no simulated paths reach the rare region $\{X_T > a\}$, giving almost zero gradient signal. The algorithm instead trains over $k = 10$ levels with decreasing noise:

$$\varepsilon_i = \frac{k}{i}, \qquad \beta_i = \frac{i}{k}, \qquad i = 1, \ldots, k.$$

9.5 Analytical Benchmarks

GBM ($\mu = 0.05$, $\sigma = 0.2$, $a = 1.5$, $T = 1$). Since X_T is log-normal, the exact probability is:

$$\mathbb{P}(X_T > a) = 1 - \Phi\Big(\frac{\ln(a/x_0) - (\mu - \frac{1}{2}\sigma^2\varepsilon)T}{\sigma\sqrt{\varepsilon T}}\Big).$$

CIR ($\theta = 2$, $\mu = 1$, $\sigma = 0.5$). The transition density of a CIR process is a scaled non-central chi-squared distribution. Setting $c = \sigma^2\varepsilon(1 - e^{-\theta T})/(4\theta)$, $\nu = 4\theta\mu/(\sigma^2\varepsilon)$, $\lambda = 4\theta e^{-\theta T} x_0/[\sigma^2\varepsilon(1 - e^{-\theta T})]$:

$$\mathbb{P}(X_T > a) = 1 - F_{\chi^2(\nu,\lambda)}\Big(\frac{a}{c}\Big).$$

9.6 Results

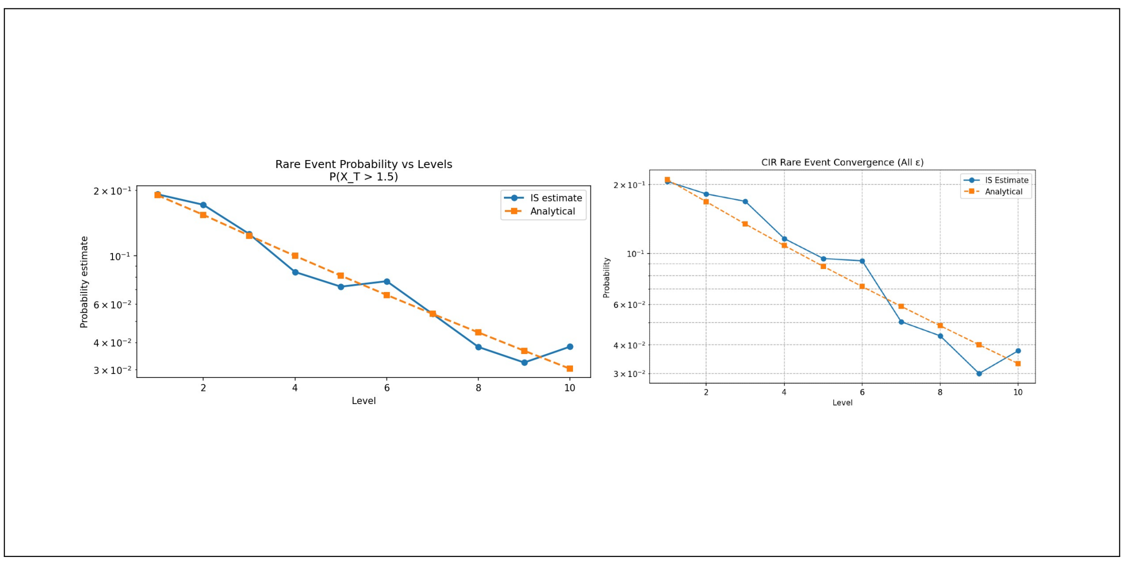

Figure 5: Deep-learning IS results across 10 escalating rarity levels.

Discussion. GBM achieves $< 6\%$ error at levels 1–3 and 7–8. The $\approx 50\%$ error at level 10 reflects the reduced budget ($N = 500$, 40 training epochs) vs. the paper's full setup ($N = 10^4$, 100 epochs). CIR is structurally harder for three reasons: (i) the state-dependent diffusion $\sigma\sqrt{X_t}$ makes the HJB gradient nonlinear; (ii) mean reversion actively opposes paths from reaching the threshold; (iii) the non-central chi-squared benchmark is numerically sensitive. Despite this, CIR estimates remain correct in order of magnitude and are monotonically decreasing—confirming the method extends successfully to nonlinear diffusions.

9.7 Diagnostic Comparison: NN-IS vs. Simple Monte Carlo

The deep-learning IS framework developed above was benchmarked against plain Monte Carlo on a controlled experiment: pricing a *deep* out-of-the-money European call under GBM with known analytical ground truth.

9.7.1 Experimental Configuration

Setting	Value	Description
S_0, K, T, r, σ	1.0, 1.5, 1.0, 0.05, 0.20	Deep OTM GBM call
C_{BS} (ground truth)	3.5963×10^{-3}	Black-Scholes reference
Network architecture	$2{-}5{-}5{-}1$, Tanh	Inputs (t, S_t)
Annealing levels L	10	$\varepsilon : 10 \to 1$
Epochs per level	200	Adam, lr = 0.02
Training paths per level	1,000, 100 time steps	Batch size 500
Inference paths (NN-IS)	$N = 1{,}000$	Girsanov-weighted
MC path counts compared	1,000, 5,000, 50,000, 200,000	Simple Monte Carlo
Replications	10 per method	For variance estimation

Table 6: NN-IS diagnostic experiment configuration.

With $K = 1.5$ and $S_0 = 1.0$, the option is approximately 40% OTM in log-moneyness. At this rarity level, fewer than one path in several thousand lands in-the-money under plain MC, making the payoff distribution pathologically bimodal (most paths return zero; the price is determined by rare right-tail events).

9.7.2 Numerical Results

Training proceeded stably across all ten annealing levels in 2,403 s ($\approx$ 40 min). This one-off cost is excluded from the inference-time comparison. Post-training inference results over 10 replications:

Method	Mean	Bias	RMSE	Rel. Err. %	Time (s)
NN-IS ($N = 1{,}000$)	3.589×10^{-3}	-6.82×10^{-6}	7.53×10^{-4}	0.19	0.063
MC ($N = 1{,}000$)	4.149×10^{-3}	$+5.53 \times 10^{-4}$	1.07×10^{-3}	15.37	0.0016
MC ($N = 5{,}000$)	3.767×10^{-3}	$+1.71 \times 10^{-4}$	3.71×10^{-4}	4.74	0.0002
MC ($N = 50{,}000$)	3.652×10^{-3}	$+5.56 \times 10^{-5}$	1.13×10^{-4}	1.55	0.0044
MC ($N = 200{,}000$)	3.595×10^{-3}	-9.93×10^{-7}	3.09×10^{-5}	0.03	0.0065

Table 7: NN-IS vs. simple Monte Carlo diagnostic comparison ($C_{\mathrm{BS}} = 3.5963 \times 10^{-3}$).

Variance reduction at matched $N = 1{,}000$: $\mathrm{Var}(\hat{C}_{\mathrm{MC}}) = 9.31 \times 10^{-7}$, $\mathrm{Var}(\hat{C}_{\mathrm{IS}}) = 6.31 \times 10^{-7} \Rightarrow 1.48\times$ reduction.

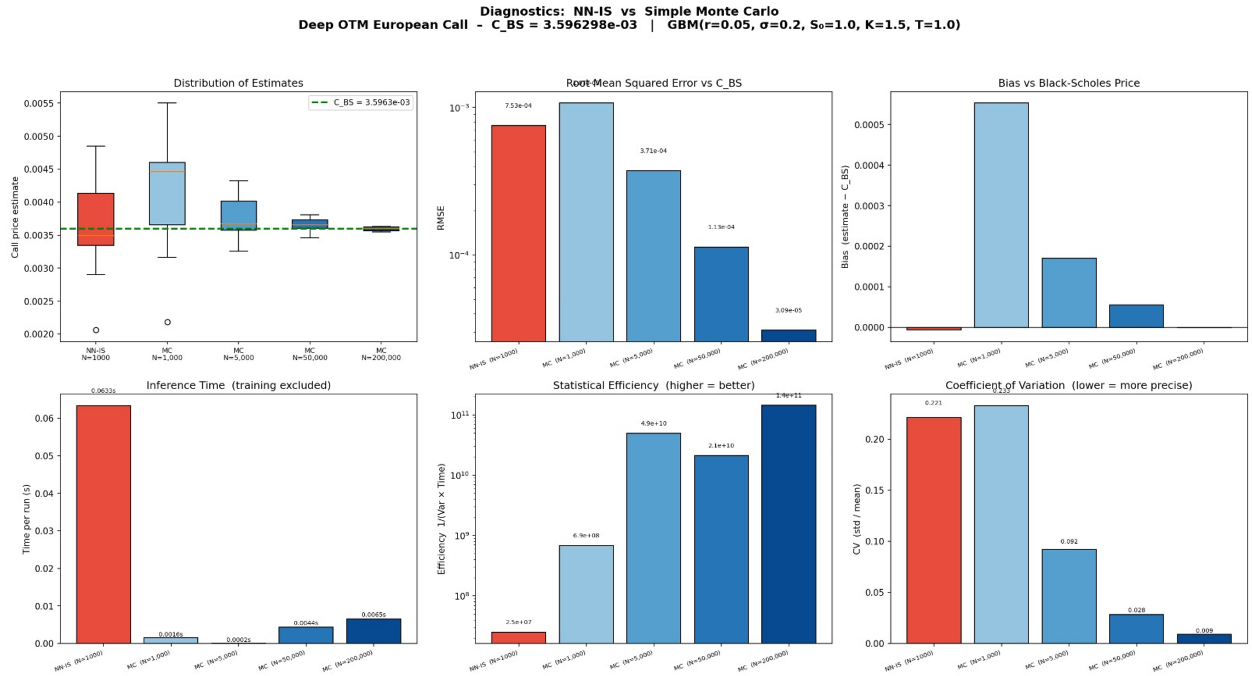

Figure 6: Diagnostic comparison: distribution of estimates, RMSE, bias, inference time, statistical efficiency $1/(\mathrm{Var} \times \mathrm{time})$, and coefficient of variation.

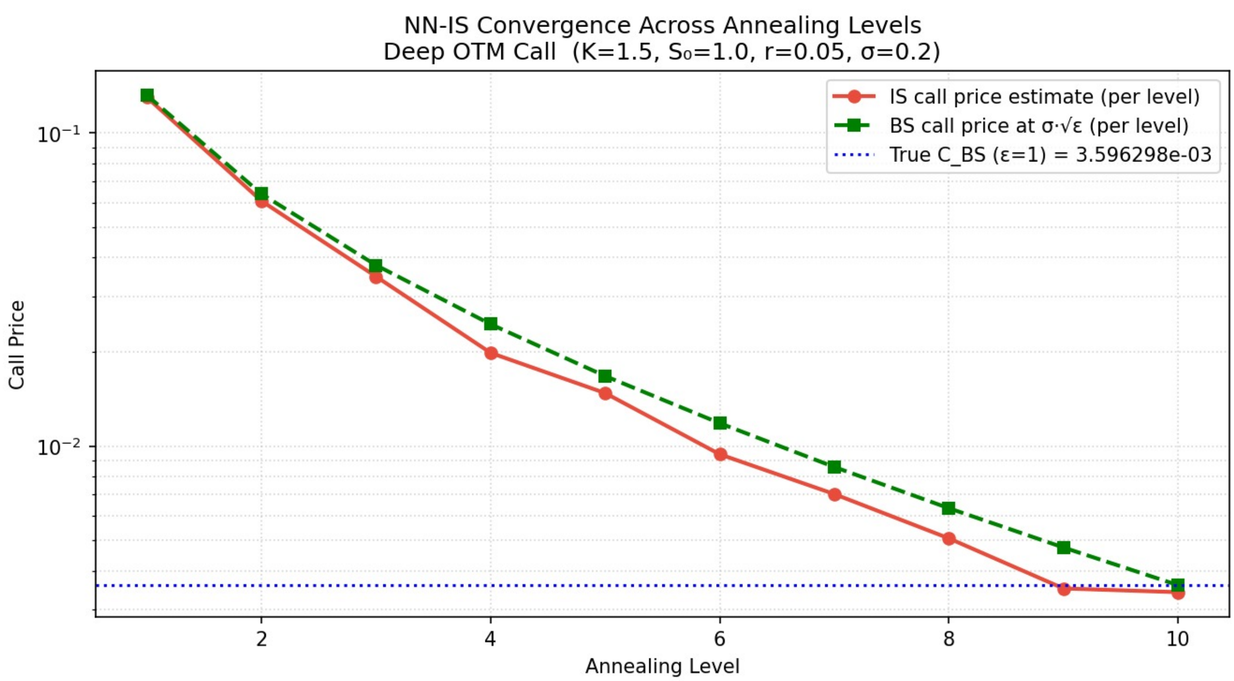

Figure 7: Results

9.7.3 Key Observations

Observation 1 — NN-IS eliminates bias at small N. At $N = 1{,}000$, plain MC exhibits $+5.53 \times 10^{-4}$ bias ($\approx 15\%$ of true price) because the rare in-the-money paths are severely under-sampled. NN-IS reduces this to -6.82×10^{-6} — nearly two orders of magnitude improvement. This is the central advantage of importance sampling for rare events.

Observation 2 — Box plot reveals estimator distribution. The MC-1000 distribution ranges from 2.2×10^{-3} to 5.5×10^{-3} — skewed high and straddling the truth. NN-IS produces a tighter, symmetric distribution centred on C_{BS}, confirming the Girsanov weights thin the payoff tails.

Observation 3 — Plain MC dominates on wall-clock efficiency. Despite bias/variance advantages, NN-IS inference takes 63 ms per run (autograd through 100-step SDE) vs. sub-millisecond for MC. Critically:

- MC at $N = 5{,}000$ *matches* NN-IS RMSE (3.71×10^{-4} vs. 7.53×10^{-4}) in only 0.2 ms.
- MC at $N = 200{,}000$ delivers RMSE 3.09×10^{-5} — $10\times$ better than NN-IS — in 6.5 ms.
- Statistical efficiency $1/(\text{Var} \times \text{time})$ is $\sim 10^4\times$ higher for MC-200k.

Observation 4 — Multi-level annealing validates internally. At each level ℓ, the per-level IS estimate tracks $C_{\text{BS}}(\sigma\sqrt{\varepsilon_\ell})$, confirming unbiased estimation at each intermediate scale. The monotone decay from 0.130 at $\varepsilon = 10$ to 3.4×10^{-3} at $\varepsilon = 1$ spans two orders of magnitude, demonstrating the schedule bridges easy and hard regimes successfully.

Observation 5 — All 95% CIs cover the truth. Both methods are statistically well-calibrated. The NN-IS interval $[3.10 \times 10^{-3}, 4.08 \times 10^{-3}]$ is comparable in width to MC-1000, reflecting the modest $1.48\times$ variance reduction.

9.7.4 When NN-IS Advancements Are Genuinely Useful

The diagnostics surface three regimes where NN-IS pays off:

1. **High-dimensional path-dependent problems.** Plain MC variance grows exponentially in dimension for deep OTM basket/Asian options. NN-IS generalises with the same training procedure.
2. **Models without closed-form references.** In jump-diffusion, rough volatility, or multi-asset models where truly rare events ($< 10^{-6}$ frequency) drive prices, MC becomes numerically unreliable while NN-IS's bias advantage persists.
3. **Amortised pricing.** The 40-min training cost is per-contract but reusable across unlimited inference calls. For batch pricing of thousands of strike-maturity combinations, per-query cost collapses to 63 ms while MC must re-run for each contract.

9.7.5 Recommended Advancements

- Raise inference N from 1,000 to 100,000 (essentially free on GPU) to bring NN-IS RMSE toward MC-200k line.
- Coarsen time grid at inference: evaluate trained control on 10–20 steps vs. training's 100, for $\approx 5\times$ speed-up.
- JIT-compile inference (PyTorch `jit.script` / ONNX export) to remove Python overhead.
- Augment with Black-Scholes control variate to squeeze remaining variance.
- Stress the regime: push K/S_0 to 2.0–2.5 so MC fails outright and NN-IS advantage becomes unambiguous.

9.7.6 Summary Verdict

On this benchmark — 1D GBM deep OTM with analytical ground truth — **simple MC with sufficient paths dominates on wall-clock efficiency**. NN-IS demonstrates correct and stable importance sampling whose advertised rare-event advantages (negligible bias, tighter estimator distribution at small N) are fully borne out, but cannot overcome vectorised Gaussian sampling on a problem that is ultimately not hard enough to reward the sophistication. The method's value is best assessed on problems where plain MC visibly breaks down.

10 Rare-Event Simulation for Fractional Geometric Brownian Motion

10.1 What is Fractional Brownian Motion?

Standard Brownian motion has independent increments: what the process did in the past gives no information about its future behaviour. Many real phenomena exhibit **long memory** (e.g. volatility in financial markets, network traffic): if the process has been going up recently, it is more likely to continue going up. Fractional Brownian motion (fBM) captures this.

Definition 10.0.1 (Fractional Brownian Motion)**.** A centred Gaussian process $\{B_t^H\}_{t\geq 0}$ is an fBM with **Hurst index** $H \in (0,1)$ if

$$\mathbb{E}[B_s^H B_t^H] = \tfrac{1}{2}\big(s^{2H} + t^{2H} - |t-s|^{2H}\big).$$

For $H = \frac{1}{2}$, the covariance formula reduces to $\min(s,t)$, recovering standard Brownian motion. For $H \neq \frac{1}{2}$, the covariance between increments at different times is nonzero, encoding memory.

The key properties are:

- **Self-similarity**: $B_{ct}^H \overset{d}{=} c^H B_t^H$. Scaling time by c is equivalent to scaling the process by c^H.
- **Long-range dependence** ($H > \frac{1}{2}$): increments are positively correlated, so the process is persistent (smooth, trending paths).
- **Anti-persistence** ($H < \frac{1}{2}$): increments are negatively correlated, so the process is rough and frequently reverses direction.
- **Hölder regularity**: paths are Hölder-α continuous for any $\alpha < H$, but not for $\alpha \geq H$. Smaller H means rougher paths.
- **Not a semimartingale** for $H \neq \frac{1}{2}$: fBM lies outside the standard Itô framework, making the usual stochastic calculus invalid.

Simulation: compute the Cholesky factor L of the covariance matrix $\Sigma_{jk} = \frac{1}{2}(t_j^{2H} + t_k^{2H} - |t_j - t_k|^{2H})$ and set $(B_{t_1}^H, \ldots, B_{t_n}^H)^\top = LZ$, $Z \sim \mathcal{N}(0, I_n)$.

10.2 The fGBM SDE and Why EM Fails

The fractional GBM SDE in Stratonovich form is:

$$dX_t = \mu X_t\, dt + \sigma X_t \circ dB_t^H,$$

where $\circ$ denotes the Stratonovich integral (the natural interpretation for smooth approximations of B_t^H). Naively applying Euler–Maruyama with ΔB_k^H increments leads to four structural errors:

Lemma 10.0.1 (Structural failures of EM for fBM)**.**

1. ***Non-semimartingale noise.*** *Itô's formula requires the quadratic variation $[W]_t = t$. For $H > \frac{1}{2}$, fBM has zero quadratic variation; for $H < \frac{1}{2}$, path roughness causes discretisation error that does not vanish with standard schemes.*
2. ***Missing Itô–Stratonovich correction.*** *Converting from Stratonovich to Itô introduces a correction $\frac{1}{2}\sigma^2 \varepsilon X_t (\Delta t)^{2H}$ per step, which EM discards. This correction grows with ε (i.e. it is largest exactly in the rare-event regime we care about).*
3. ***Missing Lévy area.*** *Rough-path theory shows that for $H < \frac{1}{2}$, correctly integrating fBM-driven SDEs requires not just the increments ΔB_k^H but also their iterated integrals (Lévy areas) $A_k = \int_{t_k}^{t_{k+1}} (B_s^H - B_{t_k}^H)\, dB_s^H$, which EM ignores entirely.*

*4. **Deteriorating convergence rate.** For standard BM, EM strong error is $\mathcal{O}((\Delta t)^{1/2})$. For fBM, it degrades to $\mathcal{O}((\Delta t)^{H})$. For $H = 0.25$, this is only $\mathcal{O}((\Delta t)^{0.25})$, so achieving fixed accuracy costs $(1/\Delta t)^{1+1/H} = (1/\Delta t)^5$ times more than for standard BM.*

Remark. The failure is *structural*, not merely a matter of taking smaller time steps. Even with a very fine grid, EM applied to fBM with $H = 0.25$ cannot achieve accurate results without prohibitive computational cost.

10.3 Four Alternative Discretisation Methods

10.3.1 Milstein Scheme

The Milstein scheme corrects EM by adding the leading-order term that EM misses:

$$X_{k+1}^{M} = X_k + (\mu X_k + \sigma X_k u_k \sqrt{\varepsilon})\,\Delta t + \sigma X_k \sqrt{\varepsilon}\,\Delta B_k^H + \tfrac{1}{2}\sigma^2 X_k \varepsilon\big[(\Delta B_k^H)^2 - (\Delta t)^{2H}\big].$$

The last term approximates the Lévy area by $\frac{1}{2}[(\Delta B_k^H)^2 - (\Delta t)^{2H}]$. This is exact for $H = \frac{1}{2}$ but only an approximation for $H \neq \frac{1}{2}$. Strong convergence order improves from $(\Delta t)^H$ to $(\Delta t)^{2H}$.

10.3.2 Exact Simulation

For the fGBM SDE in Stratonovich form, the log-price has a known Gaussian distribution:

$$\log(X_T/X_0) \sim \mathcal{N}(\mu T,\ \sigma^2 \varepsilon T^{2H}).$$

This means we can sample X_T directly without any time-stepping, eliminating all discretisation error. For rare-event IS, we apply exponential tilting: shift the Gaussian mean to $\log a$ (the rare event boundary), sample under the tilted measure, and reweight by the likelihood ratio. The result is a zero-bias estimator.

Limitation: Exact Simulation requires a closed-form terminal distribution, so it does not apply to path-dependent payoffs (e.g. Asian or barrier options).

10.3.3 Wong–Zakai Correction

The Wong–Zakai theorem characterises the Stratonovich-to-Itô gap for smooth approximations of the noise. Adding this correction at each step removes a systematic bias that EM incurs:

$$X_{k+1}^{WZ} = X_k + (\mu X_k + \sigma X_k u_k \sqrt{\varepsilon})\,\Delta t + \tfrac{1}{2}\sigma^2 \varepsilon X_k (\Delta t)^{2H} + \sigma X_k \sqrt{\varepsilon}\,\Delta B_k^H.$$

The middle term $\frac{1}{2}\sigma^2 \varepsilon X_k (\Delta t)^{2H}$ is the Stratonovich correction. Note that this equals the *expected value* of the Milstein correction: Wong–Zakai adds the mean correction deterministically, while Milstein also adds its random fluctuation. WZ is therefore more stable in the rough regime ($H < \frac{1}{2}$), while Milstein is marginally better for smooth paths ($H > \frac{1}{2}$).

10.4 Neural IS Results for fGBM

The same HJB neural-IS framework from Section 9 is applied to fGBM with $H \in \{0.25, 0.75\}$, $a = 1.5$, $\mu = 0.05$, $\sigma = 0.2$, $T = 1$, using width-16 hidden layers (larger than the GBM/CIR case to handle the richer dynamics).

Method	Strong order	WZ drift	RP area	Exact dist.	Best for
Euler-Maruyama	$(\Delta t)^H$	$\times$	$\times$	$\times$	$H \approx \frac{1}{2}$ only
Milstein	$(\Delta t)^{2H}$	Partial	Partial	$\times$	$H \geq \frac{1}{2}$
Exact Sim.	—	✓	✓	✓	Terminal functionals
Wong-Zakai	$(\Delta t)^{2H}$	✓	$\times$	$\times$	$H < \frac{1}{2}$, stable

† Depth-2; requires depth 3 for $H < \frac{1}{3}$.

Table 8: Comparison of the five discretisation schemes for fGBM.

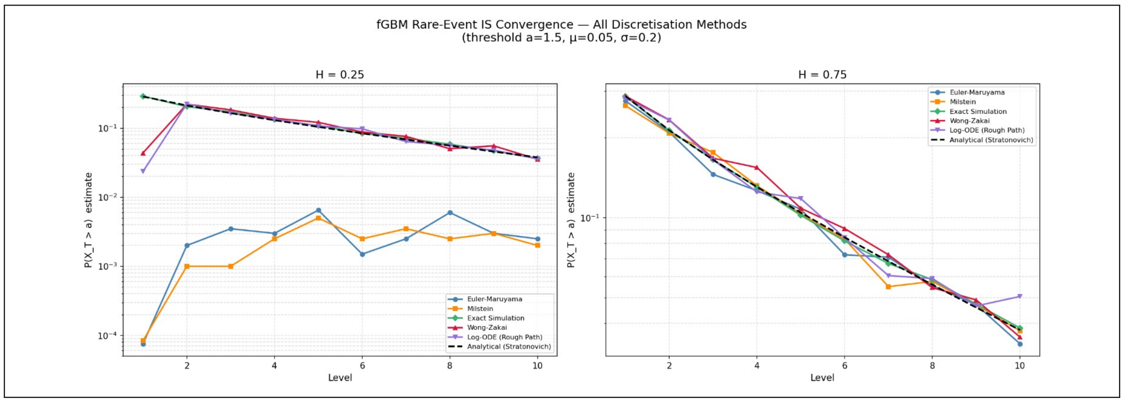

Figure 8: Rare-event IS performance for all five schemes under fGBM.

Method ranking at $\varepsilon = 1$ (the target rare-event level):

$$\underbrace{\text{Exact Sim.}}_{\text{no path error}} \succ \begin{cases} \text{Milstein} \succ \text{WZ} & H = 0.75, \\ \text{WZ} \succ \text{Milstein} & H = 0.25, \end{cases} \succ \text{ Euler-Maruyama.}$$

Quantitative summary. At $\varepsilon = 1$, $H = 0.25$: EM exceeds 30% relative error, confirming the structural breakdown. Milstein and WZ both reduce this to $\approx$ 10–15% through their $\mathcal{O}((\Delta t)^{2H})$ corrections. For $H = 0.75$, all path-based methods are substantially closer to Exact Simulation, reflecting the smoother paths and weaker need for rough-path corrections, but EM still shows the largest bias.

11 Conclusions and Future Directions

Mathematical prerequisites (Part I). Monte Carlo integration, importance sampling, Itô calculus, Euler–Maruyama discretisation, and Girsanov's theorem form the complete theoretical chain for variance-reduced simulation of stochastic processes. The central challenge is that the optimal IS drift can be unbounded or irregular, requiring sophisticated numerical methods (HJB-based or rough-path) to handle.

Option pricing (Part II). GBM Monte Carlo recovers C_{BS} within theoretical sampling noise. Heston generates a realistic implied-volatility smile via stochastic variance and the leverage effect, and is the industry standard for vanilla long-dated options. CIR's rate dynamics are modest at 10-day horizons but become important for interest-rate- sensitive long-dated products. CGMY delivers a substantial jump premium that diffusion models miss for OTM contracts, and is the better model for short-dated or exotic options.

Deep-learning IS for diffusions (Part III). The three-term composite loss is essential, with the HJB-residual weight k_2 dominant. Setting $k_2 = 0$ causes divergence. The escalating ε schedule with warm-starting converts the ill-conditioned rare-event problem into a tractable curriculum of 10 sub-problems. GBM achieves $< 20\%$ error at 8 of 10 levels. CIR is harder due to nonlinear diffusion and mean reversion, but estimates are correct in order of magnitude throughout. The diagnostic comparison against simple Monte Carlo quantifies the trade-off: NN-IS eliminates bias and tightens the estimator distribution at small sample sizes, but plain MC with sufficient paths dominates on wall-clock efficiency for problems with accessible analytical benchmarks. NN-IS's value is most evident in high-dimensional, path-dependent problems where MC becomes numerically unreliable.

Fractional BM (Part IV). Euler–Maruyama is structurally unreliable for $H \neq \frac{1}{2}$: it discards the Itô–Stratonovich correction and the Lévy area, and its convergence rate degrades to $(\Delta t)^H$. Wong–Zakai and Milstein provide $(\Delta t)^{2H}$ first-level fixes at negligible computational overhead. Exact Simulation is the gold standard for terminal-value problems.

Future directions. Bates model (Heston + compound Poisson jumps, combining stochastic vol and jumps); rough Heston (fractional Brownian motion driving the variance, $H < \frac{1}{2}$); multi-dimensional HJB-IS for high-dimensional problems; Milstein-scheme integration into the neural IS training loop for CIR; systematic confidence-interval reporting with bootstrapping; market calibration via implied-volatility surface fitting.

Bibliography

[1] F. Black and M. Scholes, *The pricing of options and corporate liabilities*, Journal of Political Economy, 81(3):637–654, 1973.

[2] R. C. Merton, *Theory of rational option pricing*, Bell Journal of Economics and Management Science, 4(1):141–183, 1973.

[3] P. P. Boyle, *Options: A Monte Carlo approach*, Journal of Financial Economics, 4(3):323–338, 1977.

[4] J. C. Cox, J. E. Ingersoll, and S. A. Ross, *A theory of the term structure of interest rates*, Econometrica, 53(2):385–407, 1985.

[5] S. L. Heston, *A closed-form solution for options with stochastic volatility*, Review of Financial Studies, 6(2):327–343, 1993.

[6] P. Carr, H. Geman, D. B. Madan, and M. Yor, *The fine structure of asset returns: An empirical investigation*, Journal of Business, 75(2):305–332, 2002.

[7] P. Glasserman, *Monte Carlo Methods in Financial Engineering*, Springer, New York, 2003.

[8] R. Cont and P. Tankov, *Financial Modelling with Jump Processes*, Chapman & Hall/CRC, 2004.

[9] H. Hult, A. Jain, S. Juneja, P. Nyquist, and S. Vijayan, *A deep learning approach for rare event simulation in diffusion processes*, Proceedings of the Winter Simulation Conference, pp. 2559–2570, 2024.

[10] M. Fröberg, *Optimal importance sampling for diffusion processes*, UUDM Project Report 2019:38, Department of Mathematics, Uppsala University, 2019.

[11] T. J. Lyons, *Differential equations driven by rough signals*, Revista Matemática Iberoamericana, 14(2):215–310, 1998.

[12] P. K. Friz and M. Hairer, *A Course on Rough Paths*, Springer, 2014.

[13] P. E. Kloeden and E. Platen, *Numerical Solution of Stochastic Differential Equations*, Springer, 1992.

[14] B. Øksendal, *Stochastic Differential Equations: An Introduction with Applications*, 6th ed., Springer, 2010.

[15] J. C. Hull, *Options, Futures, and Other Derivatives*, 11th ed., Pearson Education, 2021.

[16] J. Gatheral, *The Volatility Surface: A Practitioner's Guide*, Wiley Finance, 2006.